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Language-Driven Representation Learning for Robotics

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Abstract

Recent work in visual representation learning for robotics demonstrates the viability of learning from large video datasets of humans performing everyday tasks. Leveraging methods such as masked autoencoding and contrastive learning, these representations exhibit strong transfer to policy learning for visuomotor control. We introduce Voltron, a framework for language-driven representation learning from human videos and associated captions. The learned representations show strong gains on five downstream visuomotor control tasks spanning grasping, manipulation, and dexterous robotic control.

Introduction

This paper (arXiv:2302.12766) is titled 'Language-Driven Representation Learning for Robotics'. The work was released in 2023. The abstract reads: Recent work in visual representation learning for robotics demonstrates the viability of learning from large video datasets of humans performing everyday tasks. Leveraging methods such as masked autoencoding and contrastive learning, these representations exhibit strong transfer to policy learning for visuomotor control. We introduce Voltron, a framework for language-driven representation learning from human videos and associated captions. The learned representations show strong gains on five downstream visuomotor control tasks spanning grasping, manipulation, and dexterous robotic control. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2302.12766, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for Language-Driven Representation Learning for Robotics follows the abstract: Recent work in visual representation learning for robotics demonstrates the viability of learning from large video datasets of humans performing everyday tasks. Leveraging methods such as masked autoencoding and contrastive learning, these representations exhibit strong transfer to policy learning for visuomotor control. We introduce Voltron, a framework for language-driven representation learning from human videos and associated captions. The learned representations show strong gains on five downstream visuomotor control tasks spanning grasping, manipulation, and dexterous robotic control. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: Recent work in visual representation learning for robotics demonstrates the viability of learning from large video datasets of humans performing everyday tasks. Leveraging methods such as masked autoencoding and contrastive learning, these representations exhibit strong transfer to policy learning for visuomotor control. We introduce Voltron, a framework for language-driven representation learning from human videos and associated captions. The learned representations show strong gains on five downstream visuomotor control tasks spanning grasping, manipulation, and dexterous robotic control.

Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2302.12766 and outline directions for follow-up work.

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References

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